

LAB 1

INTRODUCTION TO PROGRAMMING

NAME:

KATARA NIKUNJ KUMAR

CLASS:

AI BTECH DIVISION B

ROLL NUMBER:

U26AI084

1.PROGRAM TO CALCULATE SIMPLE INTEREST.

INPUT:

```
#include <stdio.h>

int main() {
    float principal;
    printf("PRINCIPAL: \n");

    scanf("%f",&principal);
    float rate;

    printf("RATE \n");

    scanf("%f",&rate);

    int time ;
    printf("TIME \n");
    scanf("%d",&time);

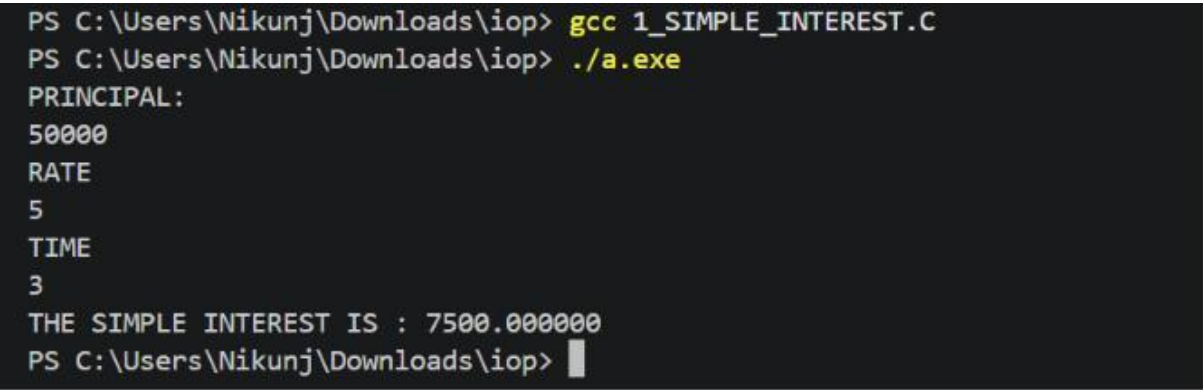
    float simple_interest;

    simple_interest=(principal*rate*time)/100;

    printf("THE SIMPLE INTEREST IS : %f \n",simple_interest);
    return 0;

}
```

OUTPUT:



```
PS C:\Users\Nikunj\Downloads\iop> gcc 1_SIMPLE_INTEREST.C
PS C:\Users\Nikunj\Downloads\iop> ./a.exe
PRINCIPAL:
50000
RATE
5
TIME
3
THE SIMPLE INTEREST IS : 7500.000000
PS C:\Users\Nikunj\Downloads\iop> █
```

2. PROGRAM TO READ MARKS OF FIVE SUBJECT OF A STUDENT AND CALCULATE TOTAL AND PERCENTAGE.

INPUT:

```
#include <stdio.h>

int main() {

    float first_subject;

    float second_subject;

    float third_subject;

    float fourth_subject;

    float fifth_subject;

    float total;

    float percentage;

    printf("enter marks \n");


    printf("first subject: \n");
    scanf("%f",&first_subject);


    printf("second subject: \n");
    scanf("%f",&second_subject);


    printf("third subject: \n");
    scanf("%f",&third_subject);


    printf("fourth subject: \n");
    scanf("%f",&fourth_subject);


    printf("fifth subject: \n");
    scanf("%f",&fifth_subject);
```

```
total=first_subject+second_subject+third_subject+fourth_subject+fifth_subject;

percentage=(total/500)*100;

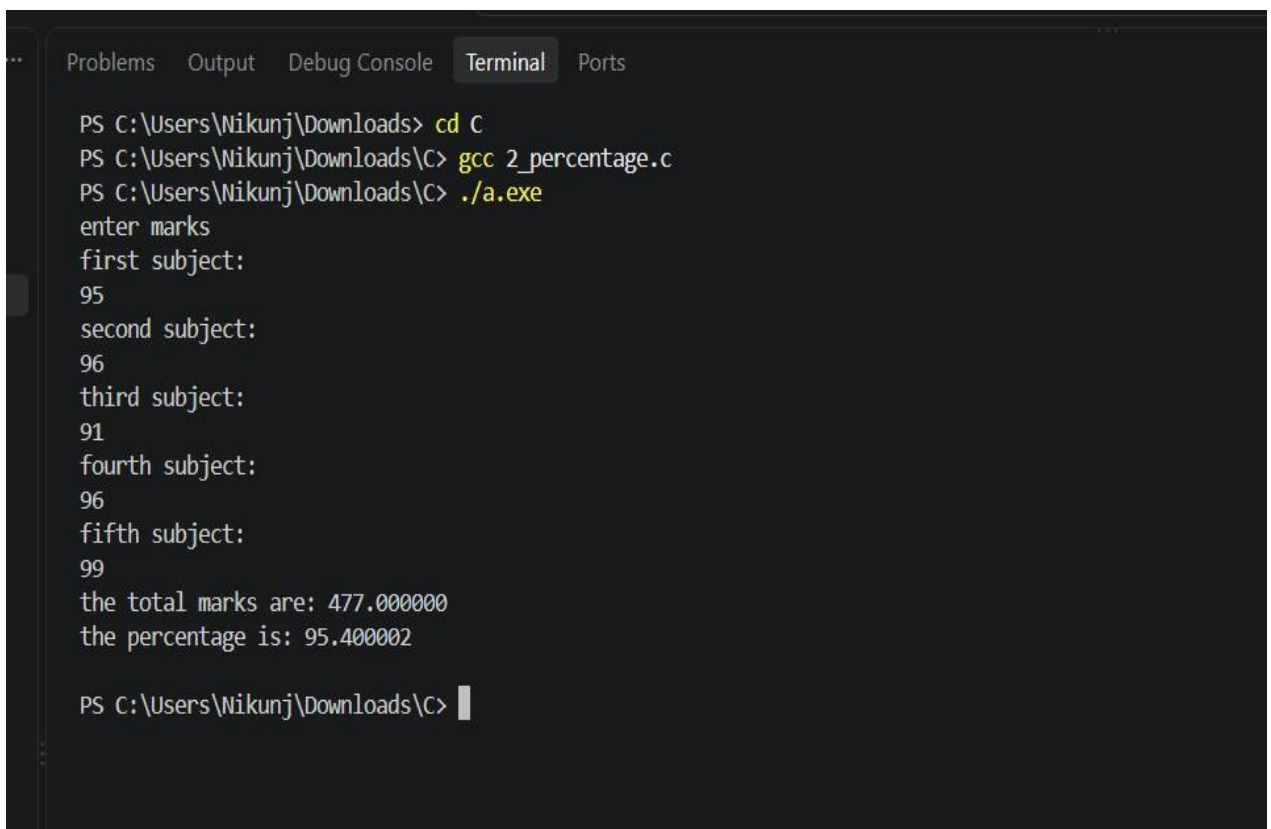
printf("the total marks are: %f \n",total);

printf("the percentage is: %f \n ",percentage);

return 0;

}
```

OUTPUT:



```
PS C:\Users\Nikunj\Downloads> cd C
PS C:\Users\Nikunj\Downloads\C> gcc 2_percentage.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
enter marks
first subject:
95
second subject:
96
third subject:
91
fourth subject:
96
fifth subject:
99
the total marks are: 477.000000
the percentage is: 95.400002

PS C:\Users\Nikunj\Downloads\C> 
```

3.PROGRAM TO CALCULATE GROSS SALARY.

INPUT:

```
#include <stdio.h>

int main() {
    float allowance;
    float basic;
    float deductions;
    float tax;
    float gross;

    printf("ENTER DETAILS \n");

    printf("ENTER BASIC SALARY: \n");
    scanf("%f",&basic);

    printf("ENTER TOTAL ALLOWANCES: \n");
    scanf("%f",&allowance);

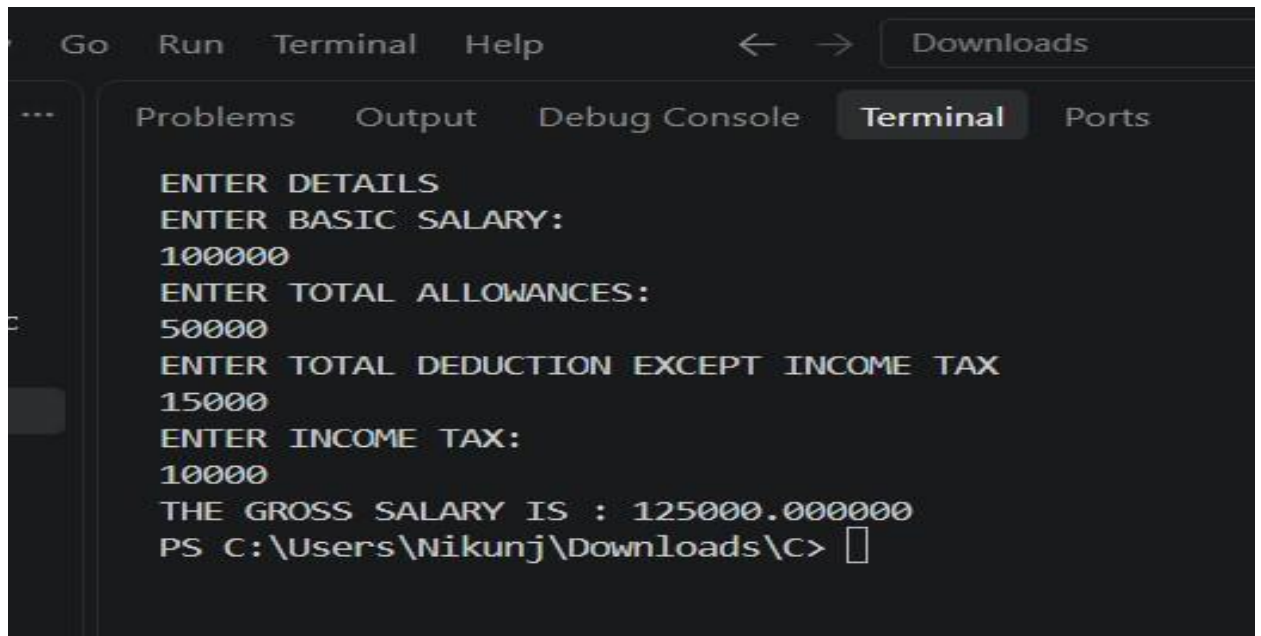
    printf("ENTER TOTAL DEDUCTION EXCEPT INCOME TAX \n");
    scanf("%f",&deductions);

    printf("ENTER INCOME TAX: \n");
    scanf("%f",&tax);

    gross=allowance+basic-tax-deductions;
    printf("THE GROSS SALARY IS : %f \n",gross);

    return 0;
}
```

OUTPUT:

A screenshot of a terminal window with a dark background. The window has a menu bar at the top with 'Go', 'Run', 'Terminal', and 'Help'. Below the menu bar are tabs for 'Problems', 'Output', 'Debug Console', 'Terminal' (which is selected and highlighted), and 'Ports'. The terminal displays the following text: 'ENTER DETAILS', 'ENTER BASIC SALARY:', '100000', 'ENTER TOTAL ALLOWANCES:', '50000', 'ENTER TOTAL DEDUCTION EXCEPT INCOME TAX', '15000', 'ENTER INCOME TAX:', '10000', 'THE GROSS SALARY IS : 125000.000000', and 'PS C:\Users\Nikunj\Downloads\C>'.

```
Go Run Terminal Help
Problems Output Debug Console Terminal Ports

ENTER DETAILS
ENTER BASIC SALARY:
100000
ENTER TOTAL ALLOWANCES:
50000
ENTER TOTAL DEDUCTION EXCEPT INCOME TAX
15000
ENTER INCOME TAX:
10000
THE GROSS SALARY IS : 125000.000000
PS C:\Users\Nikunj\Downloads\C>
```

4.PROGRAM TO CONVERT TEMPERATURE FROM FAHRENHEIT TOCENTIGRADE DEGREES.

INPUT:

```
#include <stdio.h>

int main() {

    float c;

    float f;

    printf("enter temperature in farenhiet \n :");

    scanf("%f",&f);

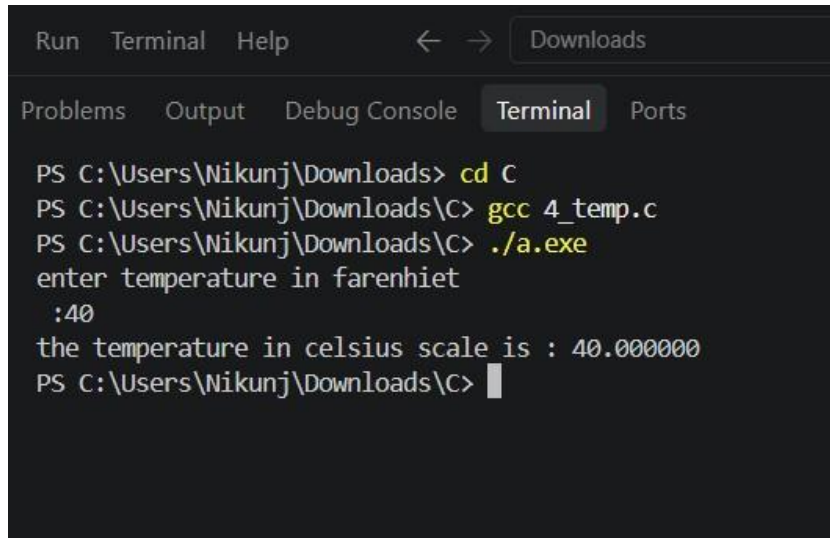
    c= (f-32.0) * 5.0/9.0;

    printf("the temperature in celsius scale is : %f \n",f);

    return 0;

}
```

OUTPUT:



```
Run Terminal Help < -> Downloads
Problems Output Debug Console Terminal Ports
PS C:\Users\Nikunj\Downloads> cd C
PS C:\Users\Nikunj\Downloads\C> gcc 4_temp.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
enter temperature in farenhiet
:40
the temperature in celsius scale is : 40.000000
PS C:\Users\Nikunj\Downloads\C> 
```


5.PROGRAM TO SWAP TOW VARIABLES USING THIRD VARIABLE.

INPUT:

```
#include <stdio.h>

int main() {

    int a;

    printf("enter a:");

    scanf("%d",&a);

    int b;

    printf("enter b:");

    scanf("%d",&b);

    printf("the values before: \n" );

    printf("%d \n",a);

    printf("%d \n",b);

    int c;

    c=a*b;

    a=c/a;

    b=c/b;

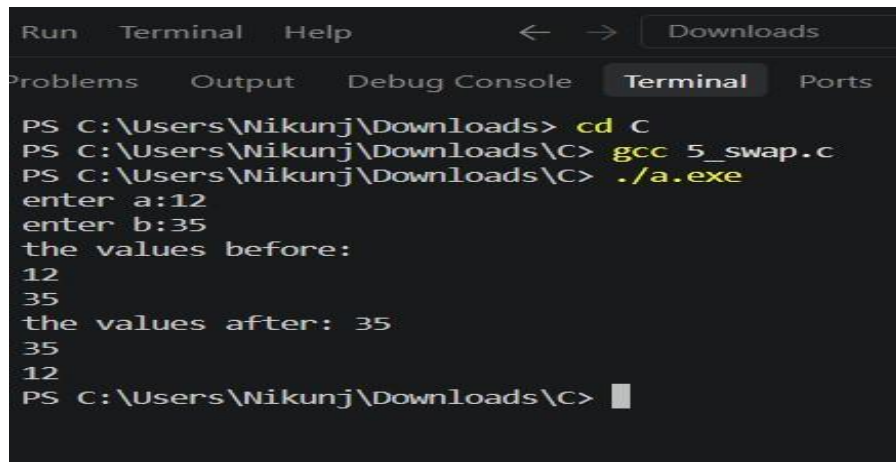
    printf("the values after: %d \n",a,b);

    printf("%d \n",a);

    printf("%d \n",b);

}
```

OUTPUT:



```
Run Terminal Help
Downloads
Problems Output Debug Console Terminal Ports
PS C:\Users\Nikunj\Downloads> cd C
PS C:\Users\Nikunj\Downloads\C> gcc 5_swap.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
enter a:12
enter b:35
the values before:
12
35
the values after: 35
35
12
PS C:\Users\Nikunj\Downloads\C>
```

6.PROGRAM TO SWAP TWO VARIABLES WITHOUT USING THIRD VARIABLE.

INPUT:

```
#include <stdio.h>

int main() {

    int a,b;

    printf("enter a,b: \n");

    scanf("%d %d",&a,&b);

    printf("values before swap:\n a=%d \n b=%d \n",a,b);

    a=a^b;

    b=a^b;

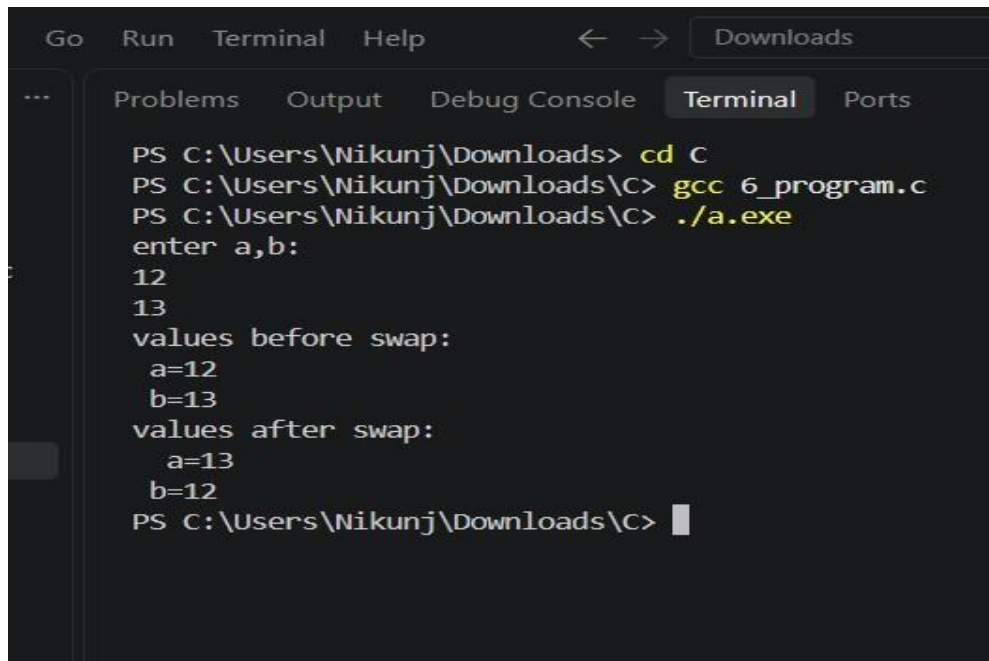
    a=a^b;

    printf("values after swap:\n a=%d \n b=%d \n",a,b);

    return 0;
```

}

OUTPUT:



```
Go Run Terminal Help Downloads
... Problems Output Debug Console Terminal Ports
PS C:\Users\Nikunj\Downloads> cd C
PS C:\Users\Nikunj\Downloads\C> gcc 6_program.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
enter a,b:
12
13
values before swap:
a=12
b=13
values after swap:
a=13
b=12
PS C:\Users\Nikunj\Downloads\C> 
```

7. PROGRAM TO CALCULATE AREA OF A TRIANGLE.

INPUT:

```
#include <stdio.h>

int main() {
    float b;
    printf("enter base:");
    scanf("%f",&b);

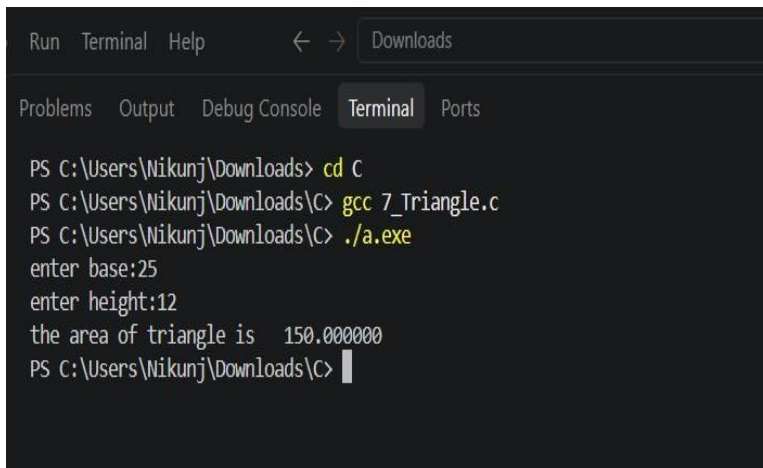
    float h;
    printf("enter height:");
    scanf("%f",&h);

    float area;
    area=0.5* b *h;

    printf("the area of triangle is % f \n",area);

    return 0;
}
```

OUTPUT:



```
Run Terminal Help  < -> Downloads

Problems Output Debug Console Terminal Ports

PS C:\Users\Nikunj\Downloads> cd C
PS C:\Users\Nikunj\Downloads\C> gcc 7_Triangle.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
enter base:25
enter height:12
the area of triangle is  150.000000
PS C:\Users\Nikunj\Downloads\C> |
```

8. EARTH TAKES A PERIOD OF REVOLUTION OF 31558150 SECONDS .WRITE A C PROGRAM TO CONVERT THIS INTO NUMBER OF DAY,HOURS AND MINUTES

INPUT:

//31558150---day hours minutes

```
#include<stdio.h>
```

```
int main() {
```

```
    printf("THE PERIOD OF REVOLUTION OF EART IS 31558150 seconds \n");
```

```
    float day;
```

```
    float hours;
```

```
    float minutes;
```

```
    minutes=31558150/60;
```

```
    hours=minutes/60;
```

```
    day=hours/24;
```

```
    printf("The total days are: %f \n",day);
```

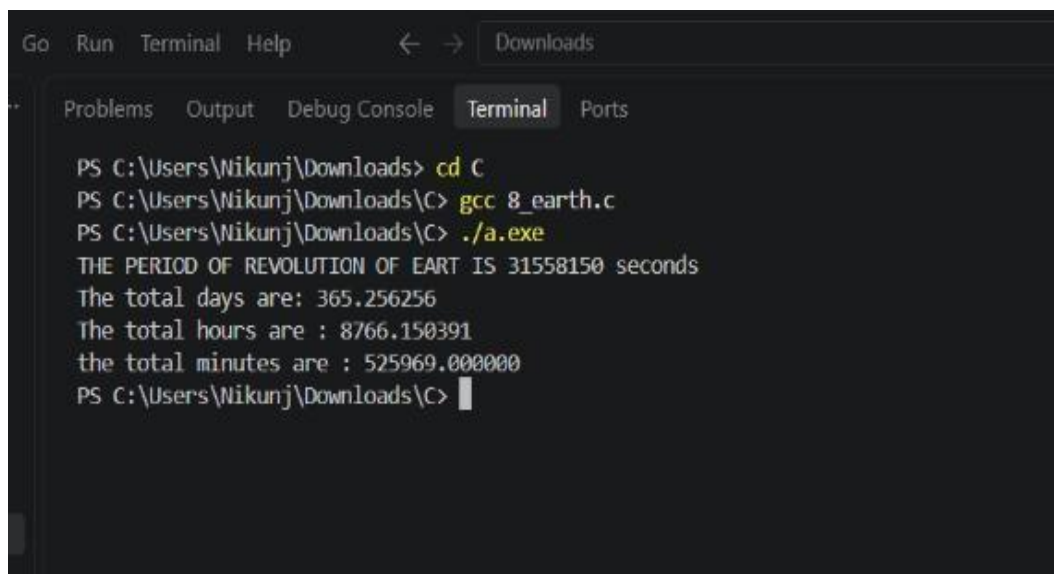
```
    printf("The total hours are : %f \n",hours);
```

```
    printf("the total minutes are : %f \n",minutes);
```

```
    return 0;
```

```
}
```

OUTPUT:



```
PS C:\Users\Wikunj\Downloads> cd C
PS C:\Users\Wikunj\Downloads\C> gcc 8_earth.c
PS C:\Users\Wikunj\Downloads\C> ./a.exe
THE PERIOD OF REVOLUTION OF EART IS 31558150 seconds
The total days are: 365.256256
The total hours are : 8766.150391
the total minutes are : 525969.000000
PS C:\Users\Wikunj\Downloads\C>
```

9. PROGRAM TO READ TIME IN HR,MIN,SEC AND CONVERT IT INTO TOTAL SECOND.

INPUT:

```
#include <stdio.h>
```

```
int main() {
```

```
    int hours;
```

```
    int minutes;
```

```
    int seconds;
```

```
    int time;
```

```
    printf("Enter current Time: \n");
```

```
    printf("enter Hours: \n");
```

```
    scanf("%d",&hours);
```

```
    printf("enter Minutes: \n");
```

```
    scanf("%d",&minutes);
```

```
    printf("enter Seconds: \n");
```

```
    scanf("%d",&seconds);
```

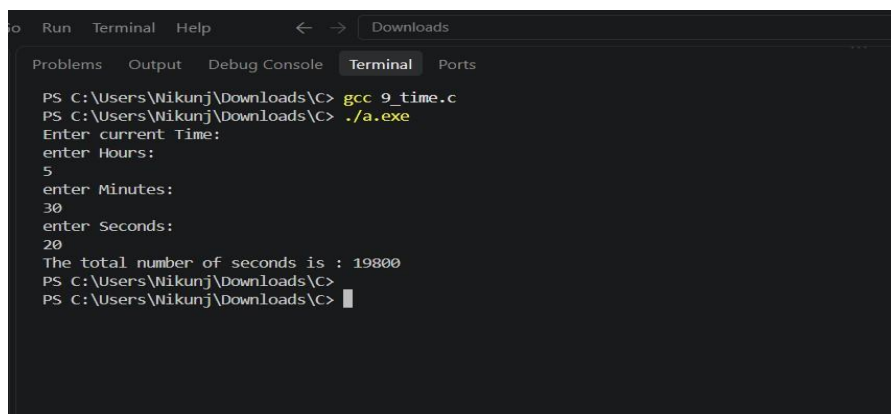
```
    time=hours*60*60+minutes*60+seconds;
```

```
    printf("The total number of seconds is : %d \n");
```

```
    return 0;
```

```
}
```

OUTPUT:



```
PS C:\Users\Nikunj\Downloads\C> gcc 9_time.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
Enter current Time:
enter Hours:
5
enter Minutes:
30
enter Seconds:
20
The total number of seconds is : 19800
PS C:\Users\Nikunj\Downloads\C>
PS C:\Users\Nikunj\Downloads\C>
```

10. WRITE A C PROGRAM TO CALCULATE THE CUT OFF MARK OF A STUDENT

INPUT:

```
#include <stdio.h>
```

```
int main(){
```

```
    int P;
```

```
    int M;
```

```
    int C;
```

```
    int E;
```

```
    int CM;
```

```
    printf("MARKS IN PHYSICS OUT OF 200: \n");
```

```
    scanf("%d",&P);
```

```
    printf("MARKS IN CHEMISTRY OUT OF 200: \n");
```

```
    scanf("%d",&C);
```

```
    printf("MARKS IN MATHS OUT OF 200: \n");
```

```
    scanf("%d",&M);
```

```
    printf("MARKS IN ENTRANCE EXAM OUT OF 100: \n");
```

```
    scanf("%d",&E);
```

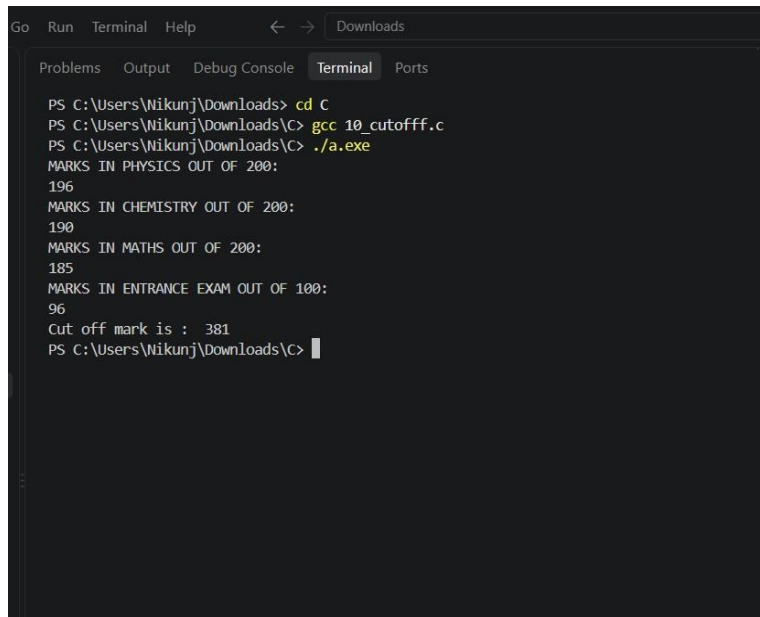
```
    CM=M/2+ P/2 + C/2 + E;
```

```
    printf("Cut off mark is : %d \n", CM);
```

```
    return 0;
```

```
}
```

OUTPUT:



```
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Problems Output Debug Console Terminal Ports

PS C:\Users\Nikunj\Downloads> cd C
PS C:\Users\Nikunj\Downloads\C> gcc 10_cutoff.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
MARKS IN PHYSICS OUT OF 200:
196
MARKS IN CHEMISTRY OUT OF 200:
190
MARKS IN MATHS OUT OF 200:
185
MARKS IN ENTRANCE EXAM OUT OF 100:
96
Cut off mark is : 381
PS C:\Users\Nikunj\Downloads\C> |
```

11.PROGRAM TO READ TOTAL SECOND AND CONVERT IT INTO TIME.

INPUT:

```
//time in seconds---time
#include <stdio.h>
int main() {
    int seconds;
    printf("enter times in seconds less than 3600s \n");
    scanf("%d",&seconds);

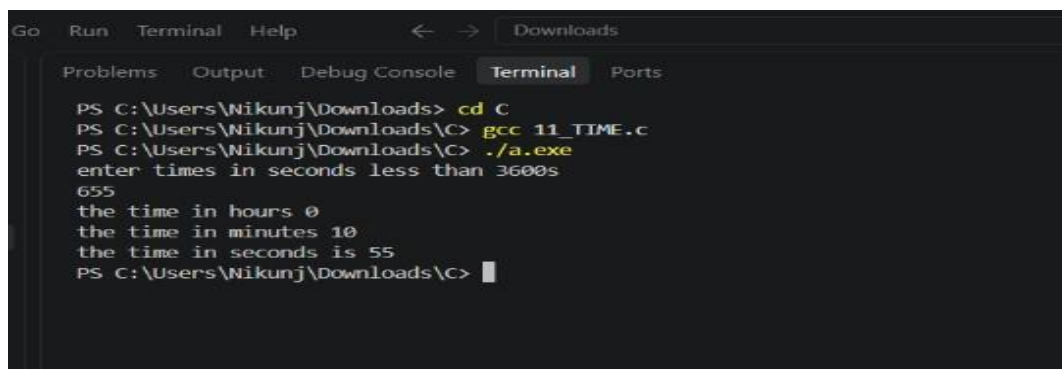
    int hour;
    int minutes ;
    int se;

    minutes=seconds/60;
    hour=minutes/60;
    se=seconds%60;

    printf("the time in hours %d \n",hour);
    printf("the time in minutes %d \n",minutes);
    printf("the time in seconds is %d \n",se);

    return 0;
}
```

OUTPUT:



```
Go Run Terminal Help Downloads
Problems Output Debug Console Terminal Ports
PS C:\Users\Nikunj\Downloads> cd C
PS C:\Users\Nikunj\Downloads\C> gcc 11_TIME.c
PS C:\Users\Nikunj\Downloads\C> ./a.exe
enter times in seconds less than 3600s
655
the time in hours 0
the time in minutes 10
the time in seconds is 55
PS C:\Users\Nikunj\Downloads\C> █
```

